SUPPORTING ADULT GUIDANCE

Letters as unknowns and variables

Formulae and Rearranging · Lesson 1. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

This opening lesson of Formulae and Rearranging separates the roles a letter can play — an unknown or a variable — and the four kinds of statement: expression, equation, formula and identity. Learners classify statements by testing values with a slider rather than by learning definitions, practise reading and writing algebraic notation, and use a tester that classifies statements they write themselves.

Driving Question: *What can a letter stand for?*

The mathematics, in plain terms

An expression is a combination of terms with no equals sign. An equation is a statement of equality that holds for particular values of its unknowns. An identity holds for all values of its variables (sometimes written with ≡). A formula expresses one quantity in terms of others, its variables. Notational conventions: $3n = 3 \times n$; ab = a × b; $n^2 = n \times n$; $n/2 = n \div 2$; a power applies only to the symbol or bracket immediately before it.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — Sorting statements by testing values

What kind of statement is each of $5y = 20$, $n + n = 2n$ and 7p – 1?

1. $5y = 20$: true for $y = 4$, false for $y = 0$.	an equation
2. $n + n = 2n$: true for $n = 3$, $n = 0$, $n = -5$ … for every value.	an identity
3. 7p – 1: no equals sign. It has a value, but makes no claim.	an expression

An equation, an identity and an expression.

Example 2 — Reading the notation when substituting

Work out $3n^2$ and $(3n)^2$ when $n = -2$.

1. $3n^2$ means $3 \times n^2$: only n is squared.	$3 \times (-2)^2 = 3 \times 4 = 12$
2. $(3n)^2$ means the whole of $3n$ is squared.	$(3 \times (-2))^2 = (-6)^2 = 36$
3. Put the negative number in brackets.	$(-2)^2 = 4$, not -4

$3n^2 = 12$ and $(3n)^2 = 36$.

When is it true?

The key question of the lesson is 'When is it true?'. Encourage learners to test statements with several values, including zero and a negative number, before deciding. If a learner calls an equation an identity, ask for a value that breaks it rather than correcting the label. One counterexample is the most powerful tool in the lesson.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	A sealed box (one hidden number) beside a stretchy rectangle (variables that change together).	~7 min
Movement	The notation decoder: evaluate notation for given values, then write long forms the short way.	~9 min
Reflection	When is it true? Slide the letter and sort statements into the four kinds.	~15 min
Creativity	Make one of each: write statements and have a tester classify them.	~9 min
Rest	One letter, four jobs: a pause on how letters change meaning with context.	~2 min
Nutrition	Intellectual: precision about what kind of claim is being made.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Reading $3n$ as '3 then n' (so $3n = 34$ when $n = 4$).
Ask the learner to write the × sign in: $3 \times n$. Then substitute.

Reading n^2 as $2n$.
Try $n = 5$: is 5^2 the same as 2×5? Ask what the small 2 means.

Squaring the coefficient as well: $3n^2$ as $(3n)^2$.
Ask which part the power is attached to, and compare both versions with $n = 2$.

Thinking a letter always stands for one fixed number.
Use the stretchy rectangle: can l be 3 and also 7? Which letters move together?

Calling a statement an identity after testing one or two values.
Ask for a test with 0 and a negative number, and ask what one counterexample would show.

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson's Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: the order of operations (multiply before you add). A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner says $2x + 3 = 11$ is an identity because it works for $x = 4$. The aim is a question that lets them find the gap themselves.

- "Does it work for $x = 0$? What do both sides give?"
- "How many values have you tried? Which ones made it true?"

- “What would an identity have to do?”
- “Can you change the statement so that it becomes an identity?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Is $y = 2x + 1$ an equation or a formula?
Both words are used. In this lesson it is treated as a formula, because y is worked out from x and both are variables. The important thing is the reasoning, not the label.

Why does the lesson use a tester rather than definitions?
Testing values gives each word a meaning the learner can check for themselves, and builds the habit of using counterexamples.

Do learners need the ≡ symbol?
It is mentioned, but not required at this stage.

What comes next?
Lesson 2 substitutes positive, negative and fractional values into expressions and formulae.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.

Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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